EE499 SENIOR DESIGN PROJECT

Report 1 — Problem & Design Requirements

Smart Home Energy Management

with Priority-Based Load Control

TEAM 16 · PME

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AGENDA

What we will cover



01

Introduction



02

Problem Statement



03

Objectives



04

Design Requirements

01

Introduction

Why household energy needs managing



Where household electricity goes

47 %

of Saudi electricity → homes

Residential share of national consumption

≈ 70 %

of home electricity → air-conditioning

Highest share worldwide

+67 %

per extra kWh above 6,000 kWh / month

18 → 30 halalas / kWh

What the single total hides

14 %

median electricity saved

with real-time, per-appliance feedback

5–15 %

extra AC energy from a dirty filter

Hidden in the total — visible per appliance

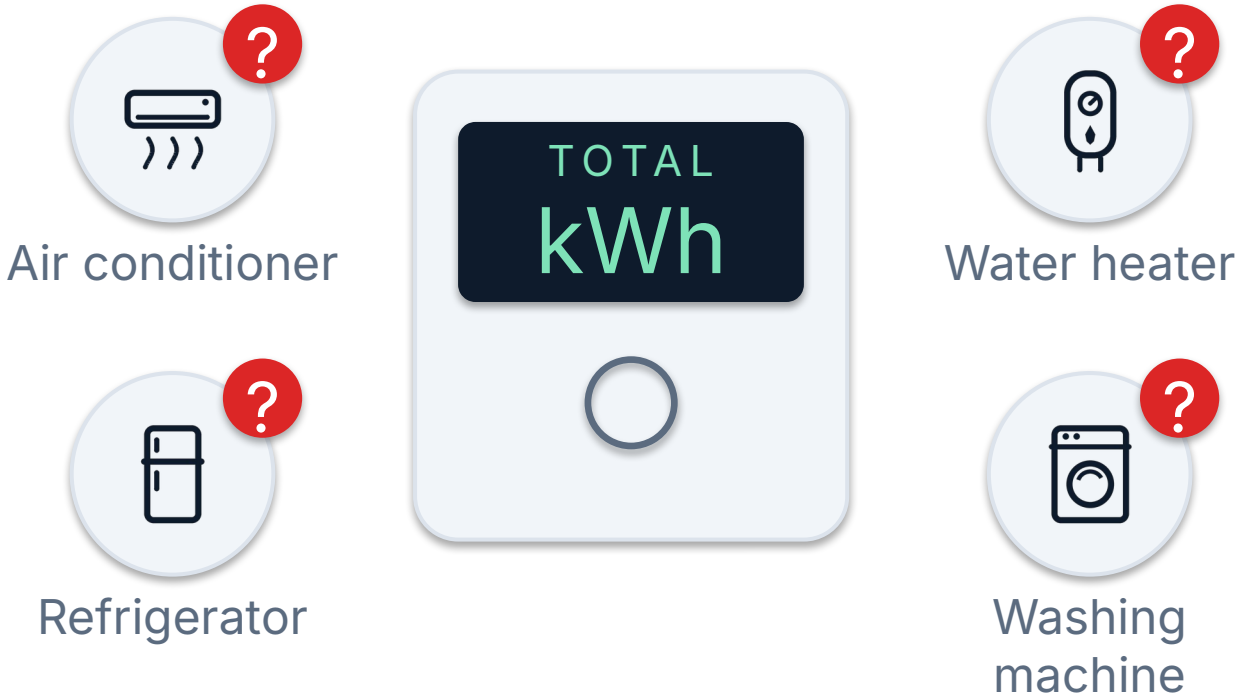
≈ 53

house fires / day in KSA

#1 cause: electrical incidents

14 %: ACEEE meta-review, 57 studies (2010) — an indication, not a promise. · 5–15 %: U.S. DOE, Energy Saver, "Maintaining your air conditioner".
53 / day: 193,437 house fires in 10 years (Civil Defense via Arab News, 2015) ÷ 3,650 days.

One meter. One number.



Not a luxury. A need.

Cost
Which appliance? → see the bill, stay in budget

Continuity
Which can wait? → essentials stay ON

Safety
Which misbehaves? → warn early, fail safe

From one total to per-appliance visibility

Conventional meter



Whole-house total only

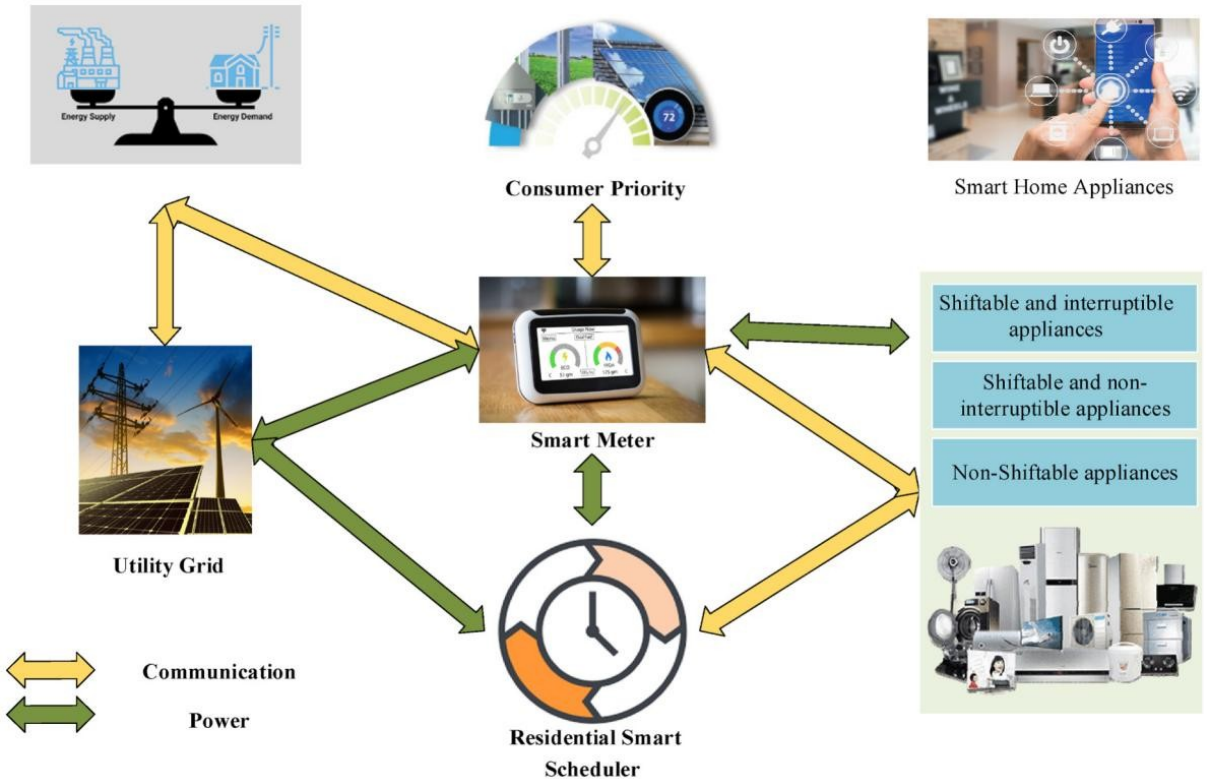
Per-appliance monitoring

- Air conditioner
- Refrigerator
- Water heater
- Washing machine

MEASURED FOR EACH APPLIANCE

- Voltage (V) and current (I)
- Power (P) and energy (kWh)
- ON / OFF status

Priority-based load management



Demand-side management concept. Reproduced from [4].

1. User sets priorities
2. Demand > limit → shed lowest
3. Essentials stay ON

Abnormal operation — a supporting function

- Deviation from baseline
- User warning
- Supports — not replaces — control

EVIDENCE

Whole-house estimate

less reliable

Direct sub-metering

more reliable

Anomaly detection on appliance data — Rashid et al., Applied Energy, 2019.

02

Problem Statement



The engineering need

What households face today

- No per-appliance visibility
- Heavy loads overlap
- Costs rise unnoticed
- Essentials not prioritised
- Abnormal behaviour hidden

The need

- Appliance-level monitoring
- Priority-based load control
- Consumption & cost management
- Essential loads kept supplied

03

Objectives

Higher-level goals · lower-level behaviours



Higher-level objectives



HLO-1

Economic

Lower consumption
Lower cost



HLO-2

Reliability

Essentials stay supplied



HLO-3

Safety & efficiency

Safer operation
Efficient use

Lower-level objectives

LLO-1
Monitor
Each load + total

LLO-2
Detect
Abnormal → warn

LLO-3
Prioritise
Shed lowest above limit

LLO-4
Budget
Energy & cost vs. budget

LLO-5
Inform
Status · cost · warnings

LLO-6
Stay safe
Fail-safe · override

04

Design Requirements

Specs · constraints · standards · assumptions · deliverables



Design specifications (1/2)

DS-01 Accuracy

±5 %

V · I · P · energy

→ LLO-1

DS-02 Update rate

≤ 3 s

Readings + status

→ LLO-1 · 5

DS-03 Abnormal detection

≥ 90 %

120 / 80 % · 5 s

→ LLO-2

DS-04 Load control

≤ 1 s

≥ 4 channels · ≥ 3 levels

→ LLO-3

Design specifications (2/2)

DS-05 Reconnection

5 s

Stable margin · highest first

→ LLO-3 · 6

DS-06 Cost & budget

±5 %

Alerts 50 · 80 · 100 %

→ LLO-4

DS-07 User warnings

≤ 5 s

7-day log · 1 h

→ LLO-2 · 5

DS-08 Fail-safe

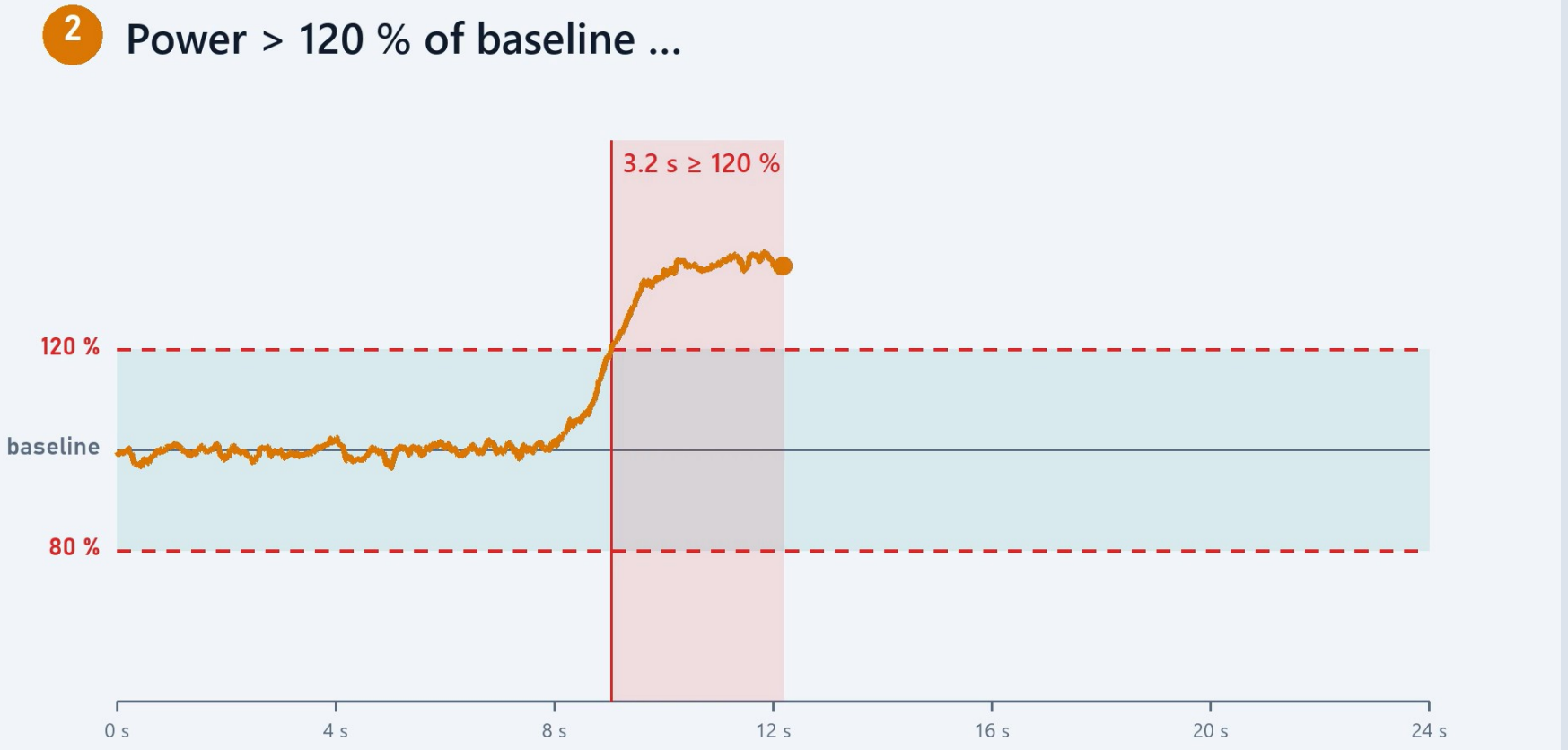
≤ 2 s

De-energize · override

→ LLO-6

REQUIRED BEHAVIOUR · DS-03 / DS-07

Abnormal-operation warning



120 / 80 %
of baseline

≥ 5 s
persistence

≥ 90 %
detection rate

≤ 5 s
to warn

Realistic constraints

C-01 · BUDGET

≤ 1,000 SAR

C-02 · SCOPE

Single-phase

C-03 · LAB POWER

≤ 2 kW

C-04 · SCHEDULE

Before the deadline

C-05 · APPLIANCES

No modification

C-06 · SAFETY

Supervised lab

Engineering standards



IEC 60730-1:2022

Automatic controls



IEC 62053-21:2020

Energy-metering accuracy



SBC 401 (2024)

Wiring & protection



Gulf Technical Regulation

Low-voltage equipment

Engineering and safety guidance — certification out of scope.

Realistic assumptions

- Normal operation at baseline
- Test loads available
- Lab + calibrated instruments
- Protected, grounded supply
- Up-to-date tariff
- Correct user inputs

Project deliverables



D-1

Prototype

D-2

Test bench

D-3

Firmware & UI

D-4

Validation results

D-5

Documentation

EE499 · TEAM 16

Thank you — questions?

Visibility. Priority. Safety. A need — not a luxury.

Team 16 · Ali Khudhayr · Mohammad Kassem · Omar Baflah

IN SUMMARY



PRIORITY CONTROL IN ACTION · DS-04 / DS-05

